

EXP :2

DATE:

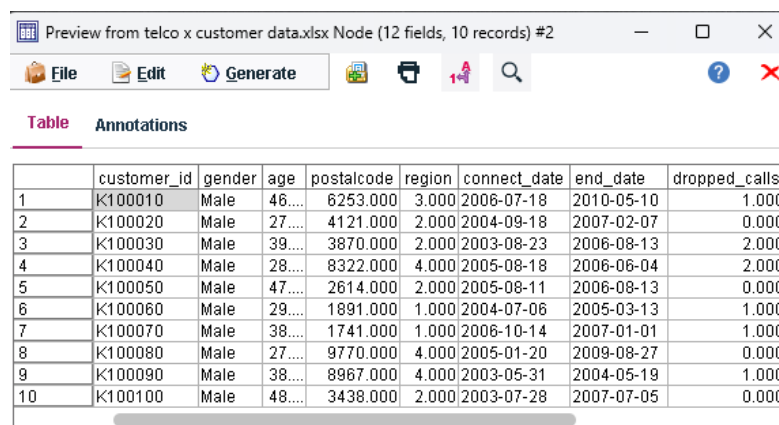
UNDERSTAND TELECOMMUNICATIONS DATA

AIM:

To Create a stream for collecting initial telecom firm data and understand the data properties using the IBM SPSS modeler.

ALGORITHM:

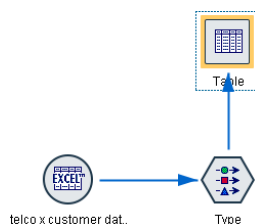
STEP 1: From the **Sources** palette, place an **Excel** node then **import the input file**, as **telco x customer data.xlsx**.



	customer_id	gender	age	postalcode	region	connect_date	end_date	dropped_calls
1	K100010	Male	46....	6253.000	3.000	2006-07-18	2010-05-10	1.000
2	K100020	Male	27....	4121.000	2.000	2004-09-18	2007-02-07	0.000
3	K100030	Male	39....	3870.000	2.000	2003-08-23	2006-08-13	2.000
4	K100040	Male	28....	8322.000	4.000	2005-08-18	2006-06-04	2.000
5	K100050	Male	47....	2614.000	2.000	2005-08-11	2006-08-13	0.000
6	K100060	Male	29....	1891.000	1.000	2004-07-06	2005-03-13	1.000
7	K100070	Male	38....	1741.000	1.000	2006-10-14	2007-01-01	1.000
8	K100080	Male	27....	9770.000	4.000	2005-01-20	2009-08-27	0.000
9	K100090	Male	38....	8967.000	4.000	2003-05-31	2004-05-19	1.000
10	K100100	Male	48....	3438.000	2.000	2003-07-28	2007-07-05	0.000

Close the Preview output window and Excel dialog box.

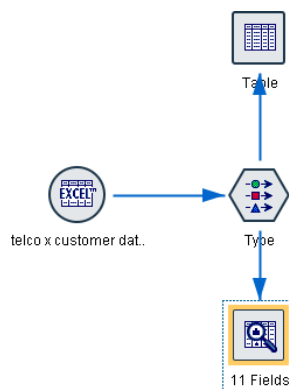
STEP 2: From the **Field Ops** palette, add a Type node and read the values in the type node. Then from the **Output** palette, add a **Table node**. Then Run the Table node.



	customer_id	gender	age	postalcode	region	connect_date	end_date	dropped_calls
1	K100010	Male	46....	6253.000	3.000	2006-07-18	2010-05-10	1.000
2	K100020	Male	27....	4121.000	2.000	2004-09-18	2007-02-07	0.000
3	K100030	Male	38....	3870.000	2.000	2003-08-23	2006-08-13	2.000
4	K100040	Male	28....	8322.000	4.000	2005-08-18	2006-06-04	2.000
5	K100050	Male	47....	2614.000	2.000	2005-08-11	2006-08-13	0.000
6	K100060	Male	29....	1891.000	1.000	2004-07-06	2005-03-13	1.000
7	K100070	Male	38....	1741.000	1.000	2006-10-14	2007-01-01	1.000
8	K100080	Male	27....	9770.000	4.000	2005-01-20	2009-08-27	0.000
9	K100090	Male	38....	8967.000	4.000	2003-05-31	2004-05-19	1.000
10	K100100	Male	48....	3438.000	2.000	2003-07-28	2007-07-05	0.000
11	K100110	Male	34....	1071.000	1.000	2005-12-11	2007-09-25	2.000
12	K100120	Male	32....	2875.000	2.000	2004-11-06	2007-10-06	0.000
13	K100130	Male	45....	8772.000	4.000	2003-12-15	2006-08-16	2.000
14	K100140	Male	29....	9016.000	4.000	2006-03-24	2008-03-13	1.000
15	K100150	Male	30....	5034.000	3.000	2005-07-28	2007-06-11	2.000
16	K100160	Male	28....	1709.000	1.000	2006-07-18	2009-09-04	9.000
17	K100170	Male	26....	1363.000	1.000	2004-06-28	2005-08-04	1.000
18	K100180	Male	37....	6455.000	3.000	2005-06-26	2006-08-06	1.000
19	K100190	Male	38....	4199.000	2.000	2004-08-06	2005-04-01	2.000
20	K100200	Male	46....	7532.000	4.000	2003-10-09	2005-08-10	0.000

Close the Table output window

STEP 3: From the **Output** palette, add a **Data Audit** node downstream from the **Type** node. Then run the **Data Audit** node.



The minimum value for **AGE** is **-1**, which is clearly an **invalid value**.

STEP 4: Edit the **Type** node Click the cell in the **Values** column, **AGE** row (where it reads [-1.0, 82.0]), and then click **Specify** the AGE Values sub-dialog box opens.

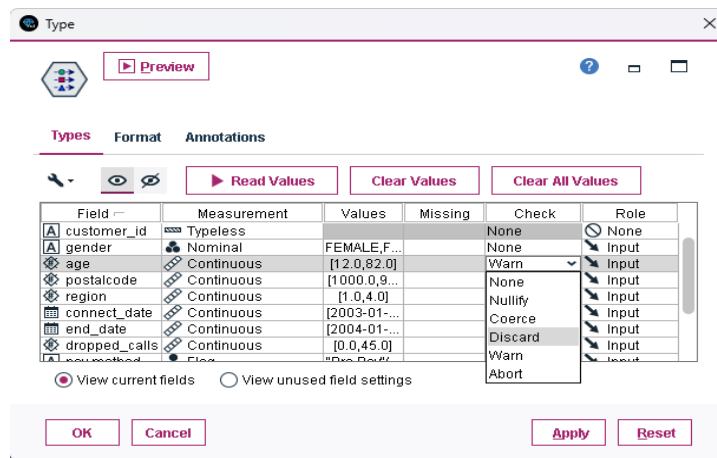
Click the **Specify** values and labels option, set then set **Lower** to **12** and **Upper** to **90**.

Close the **AGE Values** sub-dialog box.

STEP 5: Click the cell in the **Check** column, **AGE** row, and then click **Warn** from the drop-down

STEP 6: Close the **Type** dialog box. You will rerun the Data Audit node to examine the effect of specifying a valid range. Run the **Data Audit** node. The minimum AGE value is still -1, so it seems as if nothing has changed. Close the **Data Audit** output window.

STEP 7: Edit the **Type** node, click the cell in the **Check** column in the **END_DATE** row, and then set the action to **Discard**.



Close the **Type** dialog box.

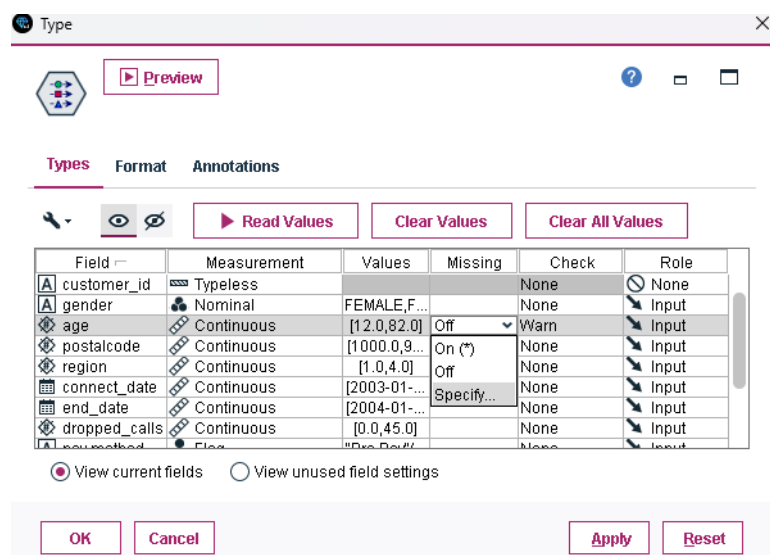
STEP 8: Run the **Table** node that is downstream from the **Type** node.

Scroll to the right so that you can view **END_DATE** and then scroll down to verify that **END_DATE** is never \$null\$.

	customer_id	gender	age	postalcode	region	connect_date	end_date	dropped_calls
1	K100010	Male	46	6253.000	3.000	2006-07-18	2010-05-10	1.000
2	K100020	Male	27	4121.000	2.000	2004-09-18	2007-02-07	0.000
3	K100030	Male	39	3870.000	2.000	2003-08-23	2006-08-13	2.000
4	K100040	Male	28	8322.000	4.000	2005-08-18	2006-06-04	2.000
5	K100050	Male	47	2614.000	2.000	2005-08-11	2006-08-13	0.000
6	K100060	Male	29	1891.000	1.000	2004-07-06	2005-03-13	1.000
7	K100070	Male	38	1741.000	1.000	2006-10-14	2007-01-01	1.000
8	K100080	Male	27	9770.000	4.000	2005-01-20	2009-08-27	0.000
9	K100090	Male	38	8967.000	4.000	2003-05-31	2004-05-19	1.000
10	K100100	Male	48	3438.000	2.000	2003-07-28	2007-07-05	0.000
11	K100110	Male	34	1071.000	1.000	2005-12-11	2007-09-25	2.000
12	K100120	Male	32	2875.000	2.000	2004-11-06	2007-10-06	0.000
13	K100130	Male	45	8772.000	4.000	2003-12-15	2006-08-16	2.000
14	K100140	Male	29	9016.000	4.000	2006-03-24	2008-03-13	1.000
15	K100150	Male	30	5034.000	3.000	2005-07-28	2007-06-11	2.000
16	K100160	Male	28	1709.000	1.000	2006-07-18	2009-09-04	9.000
17	K100170	Male	26	1363.000	1.000	2004-06-28	2005-08-04	1.000
18	K100180	Male	37	6455.000	3.000	2005-06-26	2006-08-06	1.000
19	K100190	Male	38	4199.000	2.000	2004-08-06	2005-04-01	2.000
20	K100200	Male	46	7532.000	4.000	2003-10-09	2005-08-10	0.000

Only 14,698 records are retained. Close the **Table** output window.

STEP 9: Edit the **Type** node. Click the cell in the **Missing** column, **AGE** row, and then click **Specify**.



STEP 10: Click the **Define blanks** check box to enable it. And Below **Missing values**, type **-1**.

The screenshot shows the 'age Values' dialog box. The 'Measurement' is set to 'Continuous' and 'Storage' is 'Real'. Under 'Values', 'Specify values and labels' is selected. The 'Lower' field contains '12.0' and the 'Upper' field contains '82.0'. The 'Define blanks' checkbox is checked, and the 'Missing values' field contains '-1'. The 'Check values' dropdown is set to 'Warn'. There are 'OK', 'Cancel', and 'Help' buttons at the bottom.

Close the **AGE Values** sub dialog box. **And** Close the **Type** dialog box.

Run the **Data Audit** node.

The screenshot shows the 'Data Audit' window with a table of audit results for 11 fields. The 'age' field is highlighted, showing a minimum value of 12.000. The table includes columns for Field, Sample Graph, Measurement, Min, Max, Mean, Std. Dev, Skewness, Unique, and Valid.

Field	Sample Graph	Measurement	Min	Max	Mean	Std. Dev	Skewness	Unique	Valid
gender		Nominal	--	--	--	--	--	6	31781
age		Continuous	12.000	82.000	30.327	12.858	0.829	--	31766
postalcode		Continuous	1000.000	9998.000	5512.714	2607.360	-0.006	--	31781
region		Continuous	1.000	4.000	2.672	1.057	-0.161	--	31781
connect_date		Continuous	2003-01-01	2006-12-31	--	--	--	--	31781
end_date		Continuous	2004-01-01	2010-12-29	--	--	--	--	14686
dropped_ca...		Continuous	0.000	45.000	3.190	4.195	2.152	--	31781
pay method		Flag	--	--	--	--	--	1	31781

* Indicates a multimode result * Indicates a sampled result

The minimum value for AGE is 12 instead of -1.

There are no stream messages so no out-of-range values were found.

OUTPUT:

Options

Messages

Parameters

Deployment

Execution

Globals

Search

Comments

Annotations

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Show: ☒ General messages ☐ Execution times

Message
 Starting run...
 StreamExecutionSuccess
 Stream execution complete, Elapsed=8.14 sec, CPU=4.97 sec

RESULT:

Thus, the Colleting and Understanding the telecommunication data. Program has been Executed Successfully.